PROBLEM 18

Heavy Particle on a Cone

Original statement · Problems.pdf, p. 10

Conical Surface

A heavy particle of mass m is moving without friction on a cone which is described as:

$$z^2 = x^2 + y^2$$

The frame of reference $Oxyz$ is inertial and Oz is the ascending vertical. The initial coordinates of the particle are $(a, 0, a)$ and its velocity is $\sqrt{ag}\,\mathbf{j}$. Reduce the problem to integral equations and determine the maximum and minimum value of z along the motion.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle of mass m moves without friction on the interior surface of an inverted cone given by $z^2 = x^2 + y^2$ with $z > 0$.
- **Configuration Degrees of Freedom (CDOFs):**
- Ambient space: 3 Cartesian coordinates (x, y, z).
- Holonomic geometric constraint: $\Phi(x, y, z) = x^2 + y^2 - z^2 = 0$.
- Number of CDOFs: $3 - 1 = 2$. We choose cylindrical coordinates (r, θ) with $z = r$.
- **Hypotheses:**

1. Constraint is smooth (frictionless): the reaction force $\mathbf{N}$ is strictly orthogonal to the tangent space of the cone ($W_{\mathbf{N}} = 0$).

2. Inertial frame with gravity acting along $-Oz$: $\mathbf{g} = -g\mathbf{k}$.

- **Initial State ($t = 0$):**
- Position: $(x_0, y_0, z_0) = (a, 0, a) \implies r(0) = a, \theta(0) = 0, z(0) = a$.
- Velocity: $\mathbf{v}_0 = \sqrt{ag}\,\mathbf{j} \implies v_r(0) = 0, v_\theta(0) = r(0)\dot{\theta}(0) = \sqrt{ag}, v_z(0) = 0$.
- **Parameters:** $m > 0$ [kg], $g > 0$ [m/s^2], $a > 0$ [m].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Reference Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Cylindrical Vector Basis:** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$, with:

$$\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}, \qquad \mathbf{k} = \mathbf{k}$$

- **Kinematics with Constraint $z(t) = r(t)$:**
- Position: $\mathbf{r}(t) = r\mathbf{e}_r + r\mathbf{k}$.
- Velocity: Differentiating with respect to time using Poisson's rule $\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$:

$$\mathbf{v}(t) = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{r}\mathbf{k}$$

- Modulus of velocity squared:

$$v^2 = \dot{r}^2 + r^2\dot{\theta}^2 + \dot{r}^2 = 2\dot{r}^2 + r^2\dot{\theta}^2$$

- **Change of basis (cylindrical basis).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$; the columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, so Poisson's formula gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$, as used above.
- **Surface-adapted basis.** On the cone $z = r$ (slope 45°) the unit generator is $\mathbf{e}_g = (\mathbf{e}_r + \mathbf{k})/\sqrt{2}$ and the unit normal pointing into the cone, where the particle lies, is $\mathbf{n}_{in} = (\mathbf{k} - \mathbf{e}_r)/\sqrt{2}$. The triple $\{\mathbf{e}_g, \mathbf{e}_\theta, \mathbf{n}_{in}\}$ is right-handed ($\mathbf{e}_g \times \mathbf{e}_\theta = \mathbf{n}_{in}$) and its components in $\mathcal{B}_1$ are the columns of

$$[{}_1R_2] = \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix}, \qquad \det[{}_1R_2] = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- Both matrices were checked symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

1. Conservation of Angular Momentum about Oz

Pedagogical Justification: The forces acting on the particle are gravity $\mathbf{F}_g = -mg\mathbf{k}$ and the constraint reaction $\mathbf{N}$. Gravity is parallel to Oz, so its torque about O has no z-component. The normal to the cone $f(r,z) = r - z = 0$ is $\mathbf{n} = \frac{1}{\sqrt{2}}(\mathbf{e}_r - \mathbf{k})$, which lies entirely in the meridian plane $\operatorname{span}\{\mathbf{e}_r, \mathbf{k}\}$ and intersects the Oz axis. Thus, $\mathbf{N}$ exerts zero torque about the vertical axis Oz. By the angular momentum theorem (Notes.pdf Section 4.8), L_z is an invariant of motion.

$$L_z = [\mathbf{r} \times m\mathbf{v}] \cdot \mathbf{k} = mr^2\dot\theta = \text{constant} = C_z$$

Evaluate C_z at $t = 0$:

$$C_z = m[r(0)]^2\dot\theta(0) = ma(a\dot\theta_0) = ma(\sqrt{ag}) = ma\sqrt{ga}$$

Therefore:

$$\dot{\theta}(t) = \frac{a\sqrt{ga}}{r^2(t)}$$

2. Conservation of Mechanical Energy

> **Pedagogical Justification:** Because the constraint is frictionless (bilateral holonomic), the reaction force $\mathbf{N}$ does no work ($\mathbf{N} \cdot \mathbf{v} = 0$). Gravity is conservative with potential energy $V = mgz = mgr$. Hence, total mechanical energy $E = T + V$ is strictly conserved.

Initial energy at $t = 0$:

$$T(0) = \frac{1}{2}mv_0^2 = \frac{1}{2}m(ag)$$

$$V(0) = mgz(0) = mga$$

$$E = T(0) + V(0) = \frac{1}{2}mga + mga = \frac{3}{2}mga$$

Energy at any subsequent time:

$$E = \frac{1}{2}m\left(2\dot{r}^2 + r^2\dot{\theta}^2\right) + mgr = \frac{3}{2}mga$$

Substitute $\dot{\theta} = \frac{a\sqrt{ga}}{r^2}$:

$$\frac{1}{2}m\left(2\dot{r}^2 + r^2\frac{a^3g}{r^4}\right) + mgr = \frac{3}{2}mga$$

$$m\dot{r}^2 + \frac{1}{2}mg\frac{a^3}{r^2} + mgr = \frac{3}{2}mga$$

Divide through by m:

$$\dot{r}^2 + \frac{ga^3}{2r^2} + gr = \frac{3}{2}ga$$

3. Reduction to Quadratures (Integral Equation)

Isolate $\dot{r}^2$:

$$\dot{r}^2 = \frac{3}{2}ga - gr - \frac{ga^3}{2r^2} = \frac{g}{2r^2}\left(3ar^2 - 2r^3 - a^3\right)$$

Taking the square root:

$$\frac{dr}{dt} = \pm\sqrt{\frac{g}{2}}\frac{\sqrt{-2r^3 + 3ar^2 - a^3}}{r}$$

Separating variables yields the integral equation:

$$\int_a^r \frac{r'\,dr'}{\sqrt{-2r'^3 + 3ar'^2 - a^3}} = \pm\sqrt{\frac{g}{2}}\,t$$

4. Determination of Maximum and Minimum of z (r)

> **Pedagogical Justification:** The turning points of the radial motion occur when the radial velocity vanishes, $\dot{r} = 0$, which corresponds to the roots of the cubic polynomial $P(r) = -2r^3 + 3ar^2 - a^3 = 0$.

Rewrite $P(r) = 0$:

$$2r^3 - 3ar^2 + a^3 = 0$$

Since the particle starts at $r = a$, $r = a$ must be a root:

$$2(a)^3 - 3a(a)^2 + a^3 = 2a^3 - 3a^3 + a^3 = 0$$

Divide $2r^3 - 3ar^2 + a^3$ by $(r - a)$:

$$2r^3 - 3ar^2 + a^3 = (r - a)(2r^2 - ar - a^2)$$

Factor the quadratic factor:

$$2r^2 - ar - a^2 = (r - a)(2r + a)$$

Therefore, the polynomial factors completely as:

$$2r^3 - 3ar^2 + a^3 = (r - a)^2(2r + a)$$

Substituting back into the expression for $\dot{r}^2$:

$$\dot{r}^2 = -\frac{g}{2r^2}(r - a)^2(2r + a)$$

Since $r > 0$ and $a > 0$, the factor $(2r + a) > 0$.

The square $(r - a)^2 \geq 0$, which implies:

$$\dot{r}^2 = -\frac{g(2r+a)}{2r^2}(r-a)^2 \leq 0, \qquad \forall r > 0$$

Because $\dot{r}^2$ is a squared real quantity, it must satisfy $\dot{r}^2 \geq 0$.

The only physically possible solution for all time is:

$$\dot{r}^2 = 0 \iff r(t) = a = \text{constant}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Steady Circular Orbit:** The particle is placed exactly at the minimum of the effective potential $V_{eff}(r) = mgr + \frac{ma^3 g}{2r^2}$. The initial tangential velocity $v_0 = \sqrt{ag}$ provides precisely the centripetal acceleration required to balance the inward horizontal component of the normal reaction force $\mathbf{N}$.
- **Extremal Heights:**

$$z_{min} = z_{max} = a$$

The particle maintains a constant altitude $z = a$ for all time, executing uniform circular motion with orbital frequency $\omega = \dot{\theta} = \sqrt{g/a}$.

- **Dimensional check:** $[C_z] = [m\, a\sqrt{ga}] = \text{kg}\cdot\text{m}^2\cdot\text{s}^{-1}$ (angular momentum); $[\dot{r}^2] = \text{m}^2/\text{s}^2$ and $[g\,a^3/r^2] = (\text{m}/\text{s}^2)\,\text{m} = \text{m}^2/\text{s}^2$, so every term of the energy equation has the same units.
- **Reaction on the steady orbit (sympy force balance):** with $\dot{\theta}^2 = g/a$ the horizontal acceleration is $-g\,\mathbf{e}_r$. Requiring $N\,\mathbf{n}_{in} - mg\,\mathbf{k} = -mg\,\mathbf{e}_r$ gives $\mathbf{N} = mg(\mathbf{k} - \mathbf{e}_r) = \sqrt{2}\,mg\,\mathbf{n}_{in}$, so $\|\mathbf{N}\| = \sqrt{2}\,mg$ [N], pointing into the cone, which is physically admissible (it presses the particle against the surface).

- **Stability (small radial oscillations):** the effective inertia of the r-motion is $2m$ (the kinetic energy has $2\dot{r}^2$) and $V_{eff}''(a) = 3mg/a$, hence $\omega_r^2 = 3g/(2a)$. An RK4 integration (step 10^{-4}, $g = a = 1$) of $2\ddot{r} = r\dot{\theta}^2 - g$ with $r(0) = 1.001$, $\dot{r}(0) = 0$ and $r^2\dot{\theta} = a\sqrt{ga}$ has period 5.1304 against $2\pi/\sqrt{3/2} = 5.1302$ (in units of $\sqrt{a/g}$), confirming that the circular orbit is a stable minimum of V_{eff}.

Source: Problems.pdf (Page 10)